Hallucination as Frame Misactivation: A Structural Account of LLM Error Under the Interpretive Stack Model

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Abstract

Large language model (LLM) hallucination is standardly explained as a failure of uncertainty calibration: the model generates confident output in regions of low epistemic warrant. We argue that this account, while partially correct, is structurally incomplete. Many hallucination types—including contextual confabulation, role confusion, narrative overcompletion, and cross-domain contamination —are not primarily failures of confidence calibration but failures of *frame selection*: the model activates an interpretive frame that is inappropriate for the current input context, and generates fluent, internally coherent output within that frame.

We propose a structural account of hallucination grounded in an interpretive stack model [1]: an LLM implicitly maintains a layered set of interpretive frames acquired during pretraining and fine-tuning, and hallucination occurs when the *selection rule* activates the wrong frame in response to the current input. Under this model, the apparent diversity of hallucination types can be unified within a single explanatory framework with a predictable structure.

This account has three consequences. First, it generates a classification of hallucination types by their frame-selection failure mechanism, unifying phenomena that existing taxonomies treat as distinct. Second, it reframes retrieval-augmented generation (RAG), chain-of-thought (CoT) prompting, and system prompts as practical interventions on frame activation rather than knowledge augmentation or reasoning support. Third, it identifies *Selection Stability Capacity* (SSC)—the ability to maintain the correct frame across diverse input contexts—as the structural property that debiasing interventions should target, and explains why increasing training data volume alone is insufficient to achieve it.

We derive four testable predictions distinguishing the frame-selection account from the uncertainty-calibration account, and propose mechanistic interpretability as the primary empirical route for verification. This framework does not merely reinterpret hallucination: it predicts systematic shifts in error type distribution under different

prompting conditions, retrieval strategies, and fine-tuning regimes— predictions that are falsifiable and distinct from those of the calibration account.

1 Introduction

Hallucination in large language models—the generation of fluent, confidently-stated content that is factually incorrect, contextually inappropriate, or unsupported by the input—has become a central problem in AI deployment. The standard account attributes hallucination to *epistemic failure*: the model lacks reliable uncertainty estimates and generates high-confidence output in low-confidence regions [2–4]. The canonical prescription follows directly: improve calibration through temperature scaling, uncertainty-aware training, or reinforcement learning from human feedback (RLHF).

We do not dispute that calibration failures contribute to hallucination. We argue, however, that calibration failure is insufficient to explain a significant class of hallucination phenomena. Consider the following:

- A model asked about a historical figure invents plausible- sounding but false biographical details, even when it could in principle express uncertainty.
- A model playing a specific role (e.g., a medical advisor) generates output appropriate to a different role (e.g., a general conversationalist) when the input shifts register slightly.
- A model given a document to summarize introduces content from a related but distinct document not in context.
- A model generating a narrative continues the narrative in a direction consistent with its training distribution rather than with the explicit constraints given in the prompt.

In each of these cases, the hallucinated content is not random noise but *coherent within a frame*—the model is generating output that would be appropriate if a different context, role, or knowledge base were active. The error is not that the model is uncertain; it is that the model is confidently operating within the wrong interpretive frame.

We propose that this class of hallucinations is better understood as *frame misactivation*: the selection rule that governs which interpretive frame is active has chosen an inappropriate frame for the current input, and the model generates fluent output within that frame without detecting the mismatch. This is structurally distinct from uncertainty-calibration failure and calls for a different diagnostic and remediation approach.

The present paper develops this account within the interpretive stack framework of [1], applies it to derive a classification of hallucination types, reinterprets RAG and CoT as frame-selection interventions, and derives testable predictions that distinguish the frame-selection account from the calibration account.

2 The Interpretive Stack Model

We introduce the four constructs required for the hallucination analysis. The framework is developed formally in companion work [1]; the present paper is self-contained with respect

to the definitions below.

Definition 1 (Interpretive Frame)**.** *An* interpretive frame F *is a function from inputs to outputs induced by a specific configuration of a language model's weight space. For an LLM, a frame corresponds to a computational subgraph that governs generation for a class of inputs—for example, the "medical knowledge" frame, the "narrative continuation" frame, or the "instruction following" frame.*

Operational grounding. *We do not assume that frames correspond to cleanly separable weight subsets. Rather, we adopt the operational definition from mechanistic interpretability research [9, 10]: a frame is a cluster of circuits (in the sense of [7]) whose causal contribution to output is coherent across a class of inputs. Operationally, two inputs e_1 and e_2 activate the* same frame *if activation patching from e_1 to e_2 (or vice versa) leaves output behavior qualitatively unchanged for the relevant task, and activates a* different frame *if patching produces a qualitative shift in output type or style. This definition connects the interpretive stack model to empirically tractable methods and makes the existence of distinct frames a testable claim rather than a posit.*

Remark 1 (Computational-level status of the frame definition)**.** *The activation patching criterion above is a* necessary condition *for same-frame classification, not a standalone decision procedure. In practice, applying activation patching requires a prior partition of inputs into candidate frame classes; the criterion confirms or disconfirms a proposed classification, not generates one from scratch. The definition is therefore* empirically tractable given a frame hypothesis*. The computational form of the selection rule $\mathcal{S}$—which specific attention heads, MLP layers, or residual stream directions implement context-to-frame mapping—remains open (OP2 below). The interpretive stack model is a* computational-level *theory in Marr's sense: it specifies what the system computes (frame selection) without yet fully specifying the implementing mechanism. This is a deliberate starting point, not a limitation.*

Definition 2 (Interpretive Stack)**.** *An* interpretive stack $\mathcal{L} = (F_1, F_2, \ldots, F_n)$ *is an ordered sequence of interpretive frames implicitly encoded in the model's weight space. F_1 is the most recently installed or highest-priority frame. Earlier frames are residual structures from pretraining or prior fine-tuning stages.*

Definition 3 (Selection Rule)**.** *A* selection rule $\mathcal{S} : \mathcal{E} \to \{1, \ldots, n\}$ *maps the current input context e—including prompt structure, register, distributional features, and semantic similarity to prior acquisition contexts—to an index k, activating frame F_k as the operative frame for generation.*

Definition 4 (Selection Stability Capacity (SSC))**.** *Let a* task specification T *determine a target frame F^* as the frame whose output function best approximates the correct response function for T (independently of whether the model produces hallucinated output). A model exhibits* Selection Stability Capacity *with respect to F^* and T if the selection rule $\mathcal{S}$ assigns F^* across the full distribution of inputs consistent with T. SSC failure occurs when $\mathcal{S}$ activates a frame $F_k \neq F^*$ for some input e in that distribution, producing output coherent within F_k but incorrect with respect to T.*

Note on non-circularity. *The target frame F^* is fixed by the task specification T, not by whether the output is hallucinated. Hallucination is then defined as SSC failure—output generated under $F_k \neq F^*$—rather than the reverse. This ordering breaks the apparent circularity*

between "wrong frame" and "hallucinated output": the frame is wrong because it mismatches T, not because it produces hallucination.

Remark 2 (Relation to prior work and Transformer evidence)**.** *The interpretive stack model is developed formally in [1]. The present paper applies this model to hallucination without assuming familiarity with the full formal apparatus.*

The claim that Transformers implement something like frame selection is supported by existing mechanistic interpretability findings. Induction heads [8] implement a form of context-sensitive pattern completion that behaves differently depending on the distributional context of the input—a primitive form of frame selection. Attention head specialization [7] shows that different heads implement qualitatively distinct computational roles, and that these roles can be causally isolated via activation patching. Sparse autoencoder decompositions [10] have identified monosemantic features corresponding to specific concepts, domains, and registers within a single model's activation space.

The selection rule $\mathcal{S}$ is therefore not a posited abstraction but a functional description of the mechanism by which context-sensitive attention patterns route computation through different circuit subgraphs. Its formal characterization in terms of specific architectural components is the subject of OP2.

2.1 The Selection Rule

The frame that is active at any moment is determined by a *selection rule* $\mathcal{S}$.

Definition 5 (Selection Rule)**.**

$$\mathcal{S}(e) = \arg\max_k \ \mathrm{softmax}_k\left(\frac{s(e,\, C_k)}{\tau}\right)$$

where $e \in \mathcal{E}$ is the current input context, C_k is the acquisition context of frame F_k, $s(\cdot,\cdot) : \mathcal{E} \times \mathcal{E} \to [0,1]$ is a context-similarity function, and $\tau > 0$ is a temperature parameter.

Remark 3 (Cognitive interpretation of τ)**.** *The temperature τ operationalizes context-discrimination capacity. Low τ produces sharp selection (dominant frame wins reliably); high τ flattens the distribution, increasing misactivation risk. In LLMs, τ effectively rises when input register diverges from the fine-tuning distribution, when context is long, or when chain-of-thought scaffolding is absent. Each of these maps directly to a testable hallucination trigger in Predictions 1–3: register shift (Type II), context length (Type III), and prompt formulation (Type V).*

Under this rule, the hallucination mechanism is precise: when e resembles the acquisition context C_k of an archived wrong-frame F_k more closely than it resembles C_1 (the target frame), $\mathcal{S}$ selects F_k, producing hallucination. The temperature τ models context-discrimination capacity: high τ (low discrimination) increases reactivation risk; low τ (sharp discrimination) stabilizes the target frame. This connects directly to Prediction 2 (Context-Structured Hallucination Rate): hallucination rate should increase monotonically with $s(e, C_k)$ for an archived wrong frame.

2.2 Constraints on the Frame Concept

A potential objection is that the frame concept is too expressive: if any input–output pattern can be attributed to "some frame," the account explains everything and therefore predicts nothing. We address this by specifying three constraints that make the frame concept falsifiable.

Constraint 1 (Coherence). A frame F must generate internally coherent outputs across its activation domain: if inputs e_1 and e_2 activate the same frame, their outputs should be stylistically, registrally, and semantically consistent with each other and with an identifiable training context. This rules out treating every individual output as a distinct frame.

Constraint 2 (Activation patching discriminability). Two proposed frames F_1 and F_2 are empirically distinct only if there exists a pair of inputs (e_1, e_2) such that activation patching from e_1 to e_2 produces a *qualitative* shift in output type—not merely a quantitative change in token probabilities. This gives a necessary condition for frame distinctness that is testable with existing mechanistic interpretability tools [9].

Constraint 3 (Acquisition context specificity). Each frame must be associated with an acquisition context C—a characterizable region of the training distribution—such that inputs with high distributional similarity to C preferentially activate the frame. Distributional similarity can be operationalized via any of the following: embedding proximity in the model's residual stream at an early layer, prompt-template surface overlap (vocabulary, register, formatting), or estimated fine-tuning corpus overlap (when fine-tuning data is available).

The prediction holds under any proxy satisfying the following *pairwise ordering condition*: for any two inputs e_1, e_2 and target frame F_k with acquisition context C_k,

$$\delta(e_1, C_k) > \delta(e_2, C_k) \;\Rightarrow\; P(\mathcal{S}(e_1) = k) \geq P(\mathcal{S}(e_2) = k).$$

This is a rank-preservation requirement, not a functional form requirement: the prediction is that the Spearman rank correlation between $\delta(\cdot, C_k)$ and the reactivation rate $P(\mathcal{S}(\cdot) = k)$, measured over a test set of inputs, is strictly positive. This formulation is proxy-agnostic and pre-registerable. This constraint generates the falsifiable prediction of Prediction 2 (Section 5): reactivation rate should be monotonically related to input similarity to C, not uniformly distributed.

Together, these constraints rule out the degenerate case where "frame" is just a label attached post-hoc to any observed output pattern. The taxonomy in Section 3 instantiates each type with a specific acquisition context, making the proposed frames falsifiable through Constraints 1–3.

3 A Frame-Selection Taxonomy of Hallucination

We propose that hallucination types can be classified by the mechanism of frame-selection failure. We identify five classes.

Remark 4 (Relation to existing hallucination taxonomies)**.** *Existing surveys [14, 15] classify hallucinations primarily by their output manifestation (factual, faithfulness, contextual,*

etc.) and by attributed cause (knowledge gap, attention failure, decoding artifact). The present taxonomy organizes hallucinations by their frame-selection failure mechanism*, which cuts across manifestation-based categories. A factual hallucination may be a Type I failure (distributional frame bleed) or a Type III failure (context boundary failure); a contextual hallucination may be a Type II failure (register-induced frame switch) or a Type IV failure (narrative frame overflow). The present taxonomy is therefore not a replacement for manifestation-based taxonomies but a proposed mechanistic layer beneath them, with distinct implications for debiasing targets.*

3.1 Type I: Distributional Frame Bleed

The model is presented with an input that shares surface features with a high-frequency training context, and the selection rule activates the frame associated with that training context rather than the intended frame.

Example. A question about a rare historical figure activates the "famous biography" frame, generating plausible-sounding but false biographical details that fit the statistical pattern of well-documented figures.

Mechanism. The selection rule $\mathcal{S}$ uses surface similarity to acquisition contexts as its primary signal. When the input resembles a high-frequency training context more than the low-frequency truth, the wrong frame wins.

Calibration account. The calibration account attributes this to overconfidence in a low-evidence region. The frame-selection account makes a stronger structural claim: even if the model were perfectly calibrated, it would generate the wrong content as long as the wrong frame is active, because the frame determines what counts as coherent output.

3.2 Type II: Register-Induced Frame Switch

A shift in input register (formality, domain vocabulary, persona cue) triggers a frame transition that is inappropriate for the task.

Example. A model instructed to act as a strict factual assistant generates speculative content when the user shifts to a conversational register, because the conversational register activates a "chatbot" frame with different generation norms.

Mechanism. The selection rule is sensitive to stylistic features of the input. A register shift that resembles the acquisition context of a different frame can override the intended frame even when the task instruction remains explicit.

3.3 Type III: Context Boundary Failure

The model generates content from a document or context not present in the current input window, treating the current context as continuous with a related prior context.

Example. A model asked to summarize document A introduces claims from document B, which is distributionally adjacent to document A in the model's training data and shares surface features with the current input, causing the selection rule to activate a frame whose acquisition context encompasses document B's domain.

Mechanism. The selection rule activates a frame whose acquisition context includes the adjacent document, effectively extending the context boundary beyond the current input.

3.4 Type IV: Narrative Overcompletion

The model generates output that continues a narrative trajectory consistent with its training distribution, overriding explicit constraints in the prompt.

Example. A model asked to write a story with an unusual ending instead generates a conventional ending that fits the genre template, because the narrative completion frame is more strongly activated than the "constraint following" frame.

Mechanism. Inputs with narratively structured context produce selection outcomes in which the narrative-completion frame is assigned by $\mathcal{S}$ with higher probability than the constraint-following frame, because the narrative structure more closely resembles the narrative frame's acquisition context. The prompt constraint is processed within the already-active narrative frame rather than overriding the selection outcome.

3.5 Type V: Epistemic Frame Displacement

The model responds as though it has access to information it does not have, or as though it lacks information it does have, because the active frame determines what counts as "known."

Example. A model confidently asserts outdated information because the frame active for that domain was acquired during a period when that information was correct, and the selection rule does not update to a more recent frame.

Mechanism. The selection rule activates a temporally anchored frame. Within that frame, the outdated information is correct and confident assertion is appropriate. The error is not overconfidence per se but temporal frame misalignment.

Distinction from knowledge cutoff failure. Type V is structurally distinct from ordinary knowledge cutoff failure, in which the model lacks the relevant information entirely. The distinguishing test is frame-signal injection: present the model with an explicit temporal marker (e.g., "as of 2026") alongside the query. Under a knowledge cutoff failure, the temporal marker cannot override the error because the updated information is absent from the weight space. Under Type V, the temporal marker functions as a frame-selection signal, biasing the selection rule toward a more recent acquisition context and suppressing the outdated frame—so the error *does* resolve when the frame signal is provided and *re-emerges* when it is removed. This reactivation pattern under signal removal is the signature prediction of the frame-selection account for Type V, and is not predicted by the knowledge cutoff account.

Concrete prompt pair. Consider a domain where the model's training data spans two periods with contradictory factual content (e.g., a leadership position that changed mid-training).

- **Baseline prompt:** "Who is the CEO of [company]?" The model asserts the pre-change name confidently.
- **Frame-signal prompt:** "As of early 2026, who is the CEO of [company]?" Under Type V, the temporal marker shifts the selection rule toward a later acquisition context;

the model now returns the post-change name.

- **Removal probe:** Remove the temporal marker and requery in the same conversation turn. Type V predicts reversion to the earlier name; knowledge cutoff failure predicts no change (the model had accessed the correct information and should retain it in context).

The reversion in the removal probe is the critical observation: it indicates that the correct information was available but suppressed by frame reactivation, not that it was absent.

3.6 Summary Table

Type	Name	Trigger	Mechanism
I	Distributional Bleed	Surface similarity	High-frequency frame dominance
II	Register Switch	Stylistic shift	Register-frame coupling
III	Context Boundary	Adjacent context	Context window ambiguity
IV	Narrative Overcompletion	Narrative structure	Completion prior dominance
V	Epistemic Displacement	Temporal anchoring	Temporally fixed frame

4 Reinterpreting RAG, CoT, and System Prompts

Under the frame-selection account, the major practical techniques for reducing hallucination are not primarily knowledge augmentation or reasoning support but *interventions on the selection rule*.

4.1 Retrieval-Augmented Generation (RAG)

The standard interpretation of RAG is that it provides the model with external knowledge that it lacks internally. The frame-selection account offers a complementary interpretation: RAG modifies the input context e in ways that bias the selection rule toward frames consistent with the retrieved content.

Specifically, retrieved documents carry distributional signals (vocabulary, register, domain markers) that shift the similarity between e and frame acquisition contexts. By injecting content that closely resembles the acquisition context of the target frame, RAG increases the probability that $\mathcal{S}$ selects the appropriate frame.

This interpretation makes a prediction that the knowledge augmentation account does not: RAG should reduce hallucination even when the retrieved content is semantically relevant but factually redundant—because the frame-selection effect operates through distributional similarity, not through logical entailment.

4.2 Chain-of-Thought Prompting

CoT prompting is standardly interpreted as enabling the model to perform multi-step reasoning by externalizing intermediate steps. The frame-selection account offers a structural

reinterpretation: CoT *sequentially fixes* the interpretive frame at each reasoning step, preventing frame drift between steps.

By making the reasoning process explicit in the context, CoT continuously updates the input e with the current reasoning state, biasing the selection rule toward frames consistent with the established reasoning trajectory. This reduces the probability of a frame switch mid-reasoning that would generate output inconsistent with the established conclusion.

The frame-selection account predicts that CoT should be especially effective for Type IV (narrative overcompletion) and Type II (register-induced frame switch) hallucinations, where the primary risk is frame drift during an extended generation sequence.

4.3 System Prompts and Persona Instructions

System prompts function as *prior frame injections*: by establishing a role, persona, or task frame before any user input, they bias the selection rule toward the specified frame for the duration of the interaction.

However, the frame-selection account also explains the well-known fragility of persona instructions under adversarial prompting [5, 6]: a sufficiently strong input signal—one that strongly resembles the acquisition context of a different frame—can override the system prompt's prior injection, activating the alternative frame and producing output inconsistent with the intended persona.

This is not a failure of instruction following (the model "understands" the system prompt) but a failure of SSC: the selection rule is not stable under the adversarial input.

5 Distinguishing Predictions

The frame-selection account and the calibration account make distinct empirical predictions. We identify four predictions that are tractable with current mechanistic interpretability methods.

The following table maps the five hallucination types (Section 3) to the predictions that most directly test each type's proposed mechanism.

Hallucination Type	Primary Prediction	Secondary Prediction
I: Distributional Bleed	P1 (Frame-coherent errors)	P2 (Context-structured rate)
II: Register Switch	P2 (Context-structured rate)	P4 (Circuit identity)
III: Context Boundary	P3 (RAG frame signal)	P4 (Circuit identity)
IV: Narrative Overcompletion	P1 (Frame-coherent errors)	P4 (Circuit identity)
V: Epistemic Displacement	P2 (Context-structured rate)	P4 (Circuit identity)

Additional predictions for Types I–IV. Beyond the four primary predictions below, we note one confirmable prediction per hallucination type that is tractable with existing datasets: *Type I* (Distributional Bleed) predicts that clustered hallucinated outputs will show lower perplexity *within* the misactivated frame than the base model's perplexity on the same outputs; *Type II* (Register Switch) predicts that hallucination rate increases when system-prompt register mismatches the query register (e.g., formal system prompt + casual query);

Type III (Context Boundary) predicts that RAG retrieval of frame-congruent but factually redundant content reduces hallucination more than factually novel but frame-incongruent content; *Type IV* (Narrative Overcompletion) predicts that hallucinated continuations show higher cosine similarity to the narrative frame's training distribution than to the factual frame's distribution.

5.1 Prediction 1: Frame-Coherent Error Structure

Calibration account (strongest version). The most sophisticated calibration accounts [2, 4] do not claim that hallucinated content is uniformly distributed. They allow that hallucinated content is biased toward high-frequency patterns in the training distribution (since overconfident regions are precisely the high-frequency regions). On the strongest version of the calibration account, hallucinated content should correlate with training frequency and should be detectable by entropy-based uncertainty measures.

Frame-selection account. Hallucinated content should be coherent within the misactivated frame: factually wrong but stylistically, registrally, and semantically appropriate to the wrong frame. Critically, hallucinated content should be *low-entropy within the wrong frame*—it should appear confident and internally consistent rather than uncertain— because the model is generating correctly within an incorrect frame, not generating incorrectly within a correct frame. Errors should cluster by frame type rather than by training frequency.

Test. Cluster hallucinated outputs by semantic and stylistic similarity. The frame-selection account predicts cluster structure that corresponds to identifiable training frames (biography frame, narrative frame, domain expert frame, etc.).

Measurement order. To avoid circularity (Section 2.2, Constraint 2), the test proceeds in two stages. First, candidate frames are identified using the taxonomy of Section 3 as a prior: each hallucination type specifies a candidate acquisition context from which a frame hypothesis can be derived (e.g., "this Type I error should cluster with the high-frequency biography training context"). Second, activation patching is used to confirm or disconfirm the proposed frame assignment: if patching from a confirmed biography-frame input to the hallucinated output leaves output style qualitatively unchanged, the frame assignment is supported. This two-stage procedure grounds the clustering test in Constraint 2 without presupposing the result.

5.2 Prediction 2: Context-Structured Hallucination Rate

Calibration account (strongest version). Sophisticated calibration accounts [3, 4] predict that hallucination rate should vary primarily with the model's epistemic uncertainty about the topic, as estimated by consistency across multiple samples or by probing-based uncertainty measures. On this view, varying input register or vocabulary while holding topic constant should not substantially affect hallucination rate if topic-level uncertainty is controlled—the model's epistemic state about a topic should be relatively invariant to how the question is phrased.

Frame-selection account. Hallucination rate should vary with the *distributional distance* between the input context and the target frame's acquisition context. Inputs that distributionally resemble a high-frequency alternative frame should hallucinate more than

inputs that do not, independent of topic coverage. This prediction is *orthogonal* to uncertainty: a topic the model "knows well" (low entropy) can still hallucinate if register shift activates the wrong frame.

Test. Systematically vary input register and vocabulary while holding topic constant. The frame-selection account predicts that inputs resembling alternative frame acquisition contexts will hallucinate more, even for topics with high coverage and low model uncertainty.

5.3 Prediction 3: RAG Effectiveness via Frame Signal

Knowledge augmentation account. RAG should be effective primarily when retrieved content provides factual information the model lacks.

Frame-selection account. RAG should be effective even when retrieved content is factually redundant, if it provides strong frame-selection signals (vocabulary, register, domain markers). Conversely, RAG should fail when retrieved content provides correct facts but weak frame signals.

Test. Compare RAG conditions in which retrieved content is (a) factually novel and frame-congruent, (b) factually redundant and frame-congruent, (c) factually novel but frame-incongruent. The frame-selection account predicts that condition (b) reduces hallucination more than condition (c).

5.4 Prediction 4: Circuit Persistence of Misactivated Frames

Calibration account. Hallucination-prone regions of the input space should correspond to regions of high model uncertainty, identifiable by entropy in output distributions.

Frame-selection account. Hallucination-prone inputs should activate identifiable circuits associated with the misactivated frame, detectable via activation patching or causal tracing [7, 9]. The hallucinating model should show high activation of wrong-frame circuits and low activation of correct-frame circuits.

Test. Apply mechanistic interpretability methods to compare circuit activation patterns on hallucinating vs. non-hallucinating inputs. The frame-selection account predicts that the difference is in circuit identity (which frame is active) rather than in output entropy alone.

5.5 Negative Prediction: Falsification Condition

To constitute a scientific theory rather than a post-hoc classification scheme, the frame-selection account must specify conditions under which it is falsified. We state one critical negative prediction:

Negative Prediction (Type V). If the Type V (epistemic frame displacement) account is correct, then presenting an explicit temporal marker ("as of [date]") alongside a query where the model asserts outdated information should *reduce* the assertion rate, and *removing* the temporal marker from context should restore the original assertion rate. *If neither the reduction nor the restoration occurs*, the frame-selection account of Type V is falsified for that instance, and the failure is better attributed to knowledge cutoff (absence of information) rather than frame misactivation (information present but suppressed).

The critical test is the restoration step: knowledge cutoff failure predicts no change upon marker removal (the information is simply absent from weight space); frame misactivation predicts restoration (the suppressed frame reactivates when the frame signal is removed).

5.6 Competing Hypotheses and Their Predictions

Two alternative explanations merit explicit comparison:

Recency bias hypothesis. Outputs reflect recent context tokens more than semantically appropriate earlier tokens, due to attention decay over distance. On this view, hallucination follows from recency weighting, not frame selection. *Distinguishing prediction:* recency bias predicts that hallucination rate should correlate with the distance of relevant context tokens from the generation point, independent of distributional similarity to alternative acquisition contexts. The frame-selection account predicts the opposite: hallucination should correlate with distributional similarity to alternative frames, even when task-relevant context is at the same position.

Context overwrite hypothesis. Instructions or retrieved content directly overwrite the model's prior tendency, and failures arise when overwriting is incomplete. On this view, stronger instructions should produce monotonically lower hallucination rates. *Distinguishing prediction:* context overwrite predicts that longer, more emphatic system prompts should consistently reduce hallucination. The frame-selection account predicts that prompt emphasis is insufficient when the input context strongly matches an alternative frame's acquisition context—in those cases, frame-signal injection (distributional match) should dominate instruction strength. The removal probe test (Prediction 4 above) directly distinguishes the two: context overwrite predicts that removal of a frame signal should not restore prior behavior (since the instruction remains), while frame misactivation predicts restoration.

5.7 Type V Quantification

Condition	N prompts	Assertion rate	vs. baseline	Predicted by
Baseline (e_0)	100	r_0	—	All accounts
Frame signal (e_C)	100	$r_C > r_0$	$+\delta_1$	Frame-sel., Bayesian
Signal removed (e_{rm})	100	$r_{\text{rm}} \approx r_0$	$-\delta_2$	Frame-sel. only

Table 1: Type V (epistemic frame displacement) removal probe protocol. r_0: baseline assertion rate of outdated information; r_C: assertion rate under temporal-context frame signal; r_{rm}: assertion rate after signal removal. The frame-selection account predicts $r_C - r_{\text{rm}} > \delta_2$ (restoration upon removal); the knowledge cutoff account predicts $r_C - r_{\text{rm}} \approx 0$. Preregistration requires specifying $\delta_1, \delta_2 > 0$ as multiples of within-condition standard deviation before data collection. $N = 100$ per condition yields >90% power for effect sizes ≥ 0.3 SD at $\alpha = 0.05$.

6 Implications for Debiasing

6.1 Why Confidence Calibration Is Insufficient

If hallucination is primarily a frame-selection failure, then improving confidence calibration addresses the symptom but not the cause. A perfectly calibrated model that activates the wrong frame will generate output that is confidently and coherently wrong within that frame. The calibration intervention would make the model *less certain* about its wrong output, but it would not prevent the wrong frame from being activated.

The appropriate target for debiasing is SSC: the model's ability to maintain the correct frame across the full distribution of intended inputs, including inputs that distributionally resemble alternative frames.

6.2 SSC-Targeted Interventions

SSC can be targeted by interventions that modify the selection rule rather than the content of individual frames. We identify three candidate intervention types.

Acquisition context poisoning. During fine-tuning for a target frame F^*, deliberately present training examples in contexts that distributionally resemble alternative frame acquisition contexts. This trains the selection rule to select F^* even in the presence of competing frame signals.

Contrastive frame training. Train the model to explicitly distinguish frame-appropriate from frame-inappropriate outputs for a given input, rather than only training on frame-appropriate outputs. This may strengthen the selection rule's sensitivity to frame mismatch.

Context anchoring. Modify inference-time prompting to continuously re-anchor the frame signal, reducing the probability of frame drift over long generation sequences. This is a formalization of techniques already used in practice (e.g., repeating the system prompt at regular intervals).

6.3 Implications for RAG and CoT Design

Under the frame-selection account, RAG and CoT systems should be designed not only to provide relevant content or reasoning scaffolds but to maximize frame-selection signal:

(i) Retrieved documents should be filtered not only for semantic relevance but for frame congruence—documents that strongly resemble the target frame's acquisition context.

(ii) CoT prompts should be designed to explicitly re-establish the frame at each reasoning step, not only to provide intermediate conclusions.

(iii) System prompts should be tested for SSC robustness: how strongly must an adversarial input deviate from the system prompt's frame signal before the model switches frames?

7 Relation to Prior Work

The calibration account of hallucination has been developed extensively [2–4]. The frame-selection account is complementary rather than competing: we argue that calibration failure and frame misactivation are distinct failure modes that coexist in practice, and that current research over-indexes on calibration at the expense of frame-selection analysis.

The concept of "competing experts" or "mixture of behaviors" in neural networks is not new [12,13], and recent mechanistic interpretability research has identified competing circuits in LLMs [9,10]. Our contribution is to organize these observations within a formal framework (the interpretive stack) that connects them to hallucination taxonomy and generates concrete testable predictions.

The gate function in mixture-of-experts (MoE) architectures [12, 13] implements a selection rule in mathematically explicit form: a learned function assigns each input to one or more expert modules. Frame misactivation corresponds, under this analogy, to gating error. The interpretive stack model differs from MoE in three respects. First, in standard MoE, experts are trained jointly with the gate and are well-separated by design; LLM frames are acquired sequentially across pretraining and fine-tuning phases and may overlap substantially in input distribution. Second, MoE gates are trained with explicit gating objectives; the LLM selection rule $\mathcal{S}$ is emergent and not directly optimized. Third, MoE models exhibit stable gating within their training distribution by construction; LLM SSC failures occur specifically under distributional shift relative to each frame's acquisition context. The frame misactivation account therefore generalizes MoE gating failure analysis to the sequential, multi-phase training setting of large language models.

The taxonomy of hallucination types we propose differs from existing taxonomies [14, 15] in its organizing principle: rather than classifying by output type (factual vs. faithfulness hallucination, etc.), we classify by the frame-selection mechanism responsible for the error. This distinction matters because the two taxonomies imply different debiasing targets.

The relation between fine-tuning and frame persistence is developed in [1], where fine-tuning is shown to implement frame accumulation rather than overwriting, with prior frames remaining selectable under appropriate input conditions. The present paper focuses on inference-time frame selection rather than training-time frame accumulation.

A complementary perspective comes from thinking mid-training [11], which introduces an intermediate SFT+RL training phase between pretraining and post-training. The finding that this improves post-training performance by 3.2× is consistent with the frame-selection account: models that develop structured frame-transition patterns during mid-training exhibit higher Selection Stability Capacity during post-training, and are therefore less prone to the frame-misactivation failures analyzed here. The training pipeline determines not only which frames are acquired but how robustly the selection rule can discriminate among them under distributional shift.

8 Pilot Empirical Evidence

The predictions of Section 5 require mechanistic interpretability experiments at a scale beyond the scope of this theoretical paper. We report a pilot study using representational

similarity analysis (RSA) that provides initial indirect support for the frame-selection account's central structural claim: that training establishes a stable relational geometry that is sensitive to distributional input context.

Setup. We applied RSA to three training checkpoints of Pythia-70m [16]: step 0 (random initialization), step 1000 (early training), and step 143000 (fully trained reference). Pairwise dissimilarity matrices were computed over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$, using three prompt templates averaged to reduce surface-form bias. Structural similarity was measured as the second-order RSA correlation $\rho_{\text{cog}}(A, B; \ell) = \text{Spearman}(\text{upper}(D^{A,\ell}), \text{upper}(D^{B,\ell}$

Results. Two findings are relevant to the frame-selection account.

Finding 1: Training induces stable relational geometry. ρ_{cog} between early and fully trained checkpoints increases monotonically across all layers (L6: step-0 $\rightarrow$ step-1000: $0.017 \rightarrow 0.535$), indicating that training progressively installs a stable relational structure rather than reorganizing it randomly. This is consistent with the frame-selection account's premise that trained models maintain interpretive frames as structured attractors in weight space.

Finding 2: Cross-architecture convergence. $\rho_{\text{cog}}(\text{GPT-2 trained}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\text{cog}}(\text{GPT-2 trained}, \text{Pythia-step-0}) \approx 0.10$ across all layers. Two architecturally distinct models trained on different corpora converge to similar relational geometry, while random initialization shares no such structure. This suggests that the relational geometry captured by RSA is a property of training-induced frames rather than architectural artifact—consistent with the frame-selection account's prediction that hallucination patterns should be frame-coherent (Prediction 1) and context-sensitive (Prediction 2) rather than architecture-specific.

Interpretation and limitations. These results are consistent with but do not directly test Predictions 1–4, which require activation-level evidence of frame identity during hallucination. Critically, the RSA method cannot distinguish whether distributional shift in the input *changes* which frame is selected (the frame-selection account's prediction) or merely adds noise to an otherwise fixed representation. Direct tests via the Removal Probe protocol (Section 5, Prediction 4; Table 1) remain as the primary empirical priority. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

9 Open Problems

OP1. Mechanistic identification of frames. Develop a method for identifying the boundaries of interpretive frames in an LLM's weight space via sparse autoencoders [10], activation patching, or probing. This would enable empirical validation of the five hallucination types.

OP2. Formal characterization of the selection rule. Determine the computational form of $\mathcal{S}$ in transformer architectures. Which attention heads, residual stream directions, or MLP layers implement context-to-frame mapping?

OP3. SSC measurement. Develop an operational measure of SSC for LLMs: given a target frame and an adversarial input distribution, estimate the probability of frame misactivation as a function of input distributional distance from the target frame's acquisition context.

OP4. RAG frame-signal hypothesis. Test Prediction 3 empirically: does RAG reduce hallucination via frame-signal injection independent of factual content? A controlled experiment varying frame congruence while holding factual content constant would provide the critical evidence.

OP5. CoT as sequential frame fixing. Determine whether the benefit of CoT is primarily attributable to frame stabilization rather than (or in addition to) intermediate reasoning. An ablation study comparing structured CoT (which maintains frame signal) against unstructured scratchpad (which may not) would be informative.

10 Conclusion

We have proposed a structural account of LLM hallucination as frame misactivation: the selection rule governing which interpretive frame is active in generation selects an inappropriate frame for the current input context, producing output that is fluent and coherent within the wrong frame.

This account does not replace the calibration account but complements it: calibration failure and frame misactivation are distinct failure modes that call for distinct interventions. The appropriate intervention target for frame misactivation is Selection Stability Capacity (SSC)—the model's ability to maintain the correct frame across the intended input distribution—rather than confidence calibration alone.

Under this account, RAG and CoT are reinterpreted as frame-selection interventions, and their fragility under adversarial inputs is explained as SSC failure rather than instruction-following failure. The five hallucination types we identify generate concrete, falsifiable predictions that distinguish the frame-selection account from the calibration account, providing an empirical program for validation via mechanistic interpretability.

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